

NILESH BALU

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EDUCATION

Master of Science in Mechanical Engineering

CGPA: 5.91/6.00

Expected Feb 2027

ETH Zurich, Switzerland

Bachelor of Technology (Honors) in Mechanical Engineering; Minor in Systems Engineering

CGPA: 9.69/10.00

July 2024

IIT Madras, India

- Gold medal for best academic record in B.Tech Mechanical Engineering
- Gold medal for best individual project in B.Tech Mechanical Engineering

ACADEMIC HONORS & AWARDS

- **IIT Madras Academic Prizes (2020 – 2024)**: Recipient of five distinct memorial awards and department prizes for maintaining exemplary academic performance throughout the B.Tech Mechanical Engineering program.
- **Mitacs Globalink Research Internship Award (2023)**: Selected for a 12-week internship at *Ontario Tech University*, Canada, to conduct research as part of a highly competitive global cohort.
- **JEE Advanced (2020)**: Secured an All India Rank of 1082 out of 150,000 candidates (Top 0.7% nationwide) in the Joint Entrance Examination for admission to the IITs.
- **KVPY Fellowship, SX Stream (2019)**: Awarded the *Kishore Vaigyanik Protsahan Yojana* Fellowship by the Government of India. Secured an All India Rank of 1492 out of 50,000 candidates (Top 3% nationwide).

PUBLICATIONS

1. Tashiro, N. K., Ballardini, G., **Balu, N.**, and Kuchenbecker, K. J., “Human Perception of Asymmetric Vibrations under Diverse Grip Forces,” *Work in Progress (WIP) submitted to EuroHaptics 2026*.

EXPERIENCE

• Research Intern - Dynamical Modeling

Aug 2025 - Dec 2025

Haptic Intelligence Department: Guided by Prof. Katherine J. Kuchenbecker

Max Planck Institute for Intelligent Systems

1. Modeling Asymmetric Vibrations:

- Designed a lumped-parameter model and performed system identification to simulate asymmetric vibrations of the actuator and developed feedforward controllers to track a reference acceleration profile.
- Designed a user study to investigate the effect of grip force on perceived lateral forces when asymmetric vibrations are applied to the actuator.

2. System Identification of a Novel Fingertip Haptic Actuator:

- Developed a dynamical model and performed system identification to simulate high-frequency fingertip vibrations induced by the actuator by fitting experimental data to the model.

• Research Intern - Robotics

May 2023 - Aug 2023

Mechatronic and Robotic Systems Laboratory: Guided by Prof. Scott Nogleby

Ontario Tech University, Canada

- Engineered a proof-of-concept autonomous electric wheelchair for indoor navigation, using ROS as the middleware to integrate hardware and control algorithms. Implemented GMapping for SLAM using ZED cameras, Roboteq motor controllers, and Arduinos, enabling real-time environment mapping and localization.
- Enhanced odometry accuracy through sensor fusion by employing the Unscented Kalman Filter (UKF) to combine encoder and IMU data, and achieved autonomous navigation using Adaptive Monte-Carlo Localization (AMCL) and RRT-based path planning.

• Research Intern - Mechatronics and Sensor Integration

Jul 2022 - May 2023

Rehabilitation Research and Device Development (R2D2) Laboratory: Guided by Prof. Manish Anand

IIT Madras, Chennai

- Prototyped a 3-in-1 Scooter board Wheelchair, modularly switching between scooter board, indoor wheelchair with drive unit, and outdoor wheelchair with handlebar modes.
- Designed an ergonomic watch module with an ICM42688 IMU on a custom PCB for gesture and tap speed control, and developed a wireless throttle module with PID speed control of a BLDC motor via ESP32.

STUDENT ASSISTANT (HiWi) POSITIONS

• RobotX Robotics Summer School 2026

Jan 2026 - Jun 2026

- Developed a high-fidelity MuJoCo simulation for the Unitree A2, integrating a multi-modal sensor suite including IMU, proprioception, 3D LiDAR, and RGB-D cameras.
- Developed a policy for flat terrain locomotion using Unitree-RL-MJLab, leveraging MuJoCo physics to optimize robust PPO policies for the Unitree A2.

- **Programming for Robotics - Introduction to ROS** Jan 2026 - Feb 2026
– Designed and implemented course assignments focused on ROS 2 navigation and perception, providing technical mentorship to students during exercise sessions.
- **RobotX Robotics Summer School 2025** Jan 2025 - Jul 2025
– Ported the legacy software stack from *ROS* to *ROS 2*, implementing a robust low-level controller interface and a real-time object detection/localization node for autonomous artifact identification.

PROJECTS

- **Hand Pose Estimation for Digital Surgical Training and Assessment** Feb 2025 - Jul 2025
Semester Thesis: Guided by Prof. Dr. Mirko Meboldt
– Developed a real-time hand pose estimation pipeline for surgical twins using stereo vision and deep learning; deployed WiLoR on Jetson Orin to extract 2.5D keypoints and triangulate them into 3D coordinates using optimization. Improved accuracy and consistency via Kalman filtering and outlier detection, achieving sub-centimeter error.
– Validated the pipeline using ArUco markers, infrared motion capture, and blob marker techniques, demonstrating its suitability for surgical training and performance analysis.
- **Wearable Hand Assistive Device** Aug 2023 - Aug 2024
Bachelor's Thesis Project: Guided by Prof. Manish Anand
– Developed a proof-of-concept assistive hand device providing up to 10N grasping force for daily activities. Employed an *under-actuated tendon mechanism* to flex digits, reducing weight and form factor while improving compliance.
– Implemented EMG sensor-based intent detection to detect user intention and automate the flexion and extension motion of the device. Offered 2 modes of operation: *cylindrical grasp* and *lateral pinch*.
- **On the Parallelization of Seam Carving** Mar 2024 - May 2024
Course Project: Parallel Scientific Computing
– Presented an implementation of the *Seam Carving* operation that enables content-aware resizing of images. Extended the use of this operator as an object removal tool.
– Demonstrated parallelism on *OpenMP* (*Open Multi-Processing*), *MPI* (*Message Passing Interface*), and *OpenACC* (*Open Accelerators*) frameworks and compared performance evaluation metrics such as *speed-up*.
- **Numerical Analysis of Fluid Flow in a Pipe with Annular Fins** Mar 2023 - Apr 2023
Course Project: Computational Heat and Fluid Flow
– Developed a numerical solver in C++ for simulating fluid flow through a pipe with annular fins by utilizing the axisymmetric coordinate system, *Upwind Differencing Scheme (UDS)* for discretization, and the *tri-diagonal matrix algorithm (TDMA)* for solving the system of equations.
– Analyzed the impact of variations in fin parameters (length, thickness, and spacing) on fin effectiveness, noting increased effectiveness with the augmentation of these parameters.
- **Simulation of a four-tank system using Model Predictive Control** Oct 2022 - Nov 2022
Course Project: Modern Control Theory
– Developed, implemented, and fine-tuned a *Model Predictive Controller (MPC)* using Matlab for a four-tank system.
– Modeled equations that describe the system, employed the *Kalman filter* and the *Particle filter* as state estimators, and investigated the performance of the two estimators.

SKILLS

- **Analytical Skills:** Laplace Analysis, Fourier Analysis, Principal Component Analysis, Linear Estimation Theory, Model Predictive Control, Finite Volume Method, Finite Element Analysis
- **Software and Programming Languages:** Matlab and Simulink, ROS2, ROS, SolidWorks, Autodesk AutoCAD, Autodesk Fusion 360, Arduino IDE, Python, C, C++, MS Office, Latex, VESC tool, Autodesk Eagle

KEY COURSES TAKEN

- **Robotics, Systems and Control:** Model Predictive Control*, Optimal Control, Modern Control Theory, Probabilistic Artificial Intelligence, Integer Optimization, Image Analysis and Computer Vision, Robot Dynamics, Nonlinear Dynamics and Chaos I, Computational Models of Motion. Multivariate Data Analysis for Process Modelling, Signal Processing, Parallel Scientific Computing.
- **Design:** Kinematics and Dynamics of Machinery, Mechanics of Human Movement, Theory of Vibrations, Fundamentals of Finite Element Analysis, Optimization Methods in Mechanical Design.

*Ongoing